

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter Examination – 2023 Course: B. Tech. Branch: Mech. Engg/Mech. Engg. (Sandwich) Semester :VI Subject Code & Name: BTMEC601_Y19 Manufacturing Processes-II Max Marks: 60 Date:16-01-2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Define abrasive machining and explain its importance with diagram	CO1	6
B)	What are the various design considerations for grinding?	CO1	6
C)	Write a note on Truing and Shaping of Grinding Wheels.	CO1	6
Q.2	Solve Any Two of the following.		12
A)	Define metal cutting and draw a schematic diagram showing the workpiece, cutting tool, and chip.	CO2	6
B)	Explain the back rake angle, side rake angle, positive rake angle, and negative rake angle with the help of a schematic diagram	CO2	6
C)	Show all the angles and nose radius of a single point cutting tool on the different views of a single point cutting tool.	CO2	6
Q. 3	Solve Any Two of the following.		12
A)	What are the forms of wear in metal cutting? Explain in detail	CO3	6
B)	Explain the basic requirements of cutting tool materials.	CO3	6
C)	Explain powder metallurgy w.r.t. the capabilities, salient features, and the most commonly used materials	CO3	6
Q.4	Solve Any Two of the following.		12
A)	Explain Methods of Powder Production	CO4	6
B)	List and describe the functions of additives in ceramics.	CO4	6
C)	Explain Finishing Operations in ceramics	CO4	6
Q. 5	Solve Any Two of the following.		12
A)	Describe the plastic extrusion process with diagram	CO5	6
B)	Describe injection Moulding	CO5	6
C)	Write a note on Production of Polymer Reinforcing Fibers	CO5	6
	*** End ***		